EXPERIMENT ON PACKET CAPTURE TOOL: WIRESHARK

AIM:

To understand the feature of wireshark as a Packet capture tool and understood encapsulation of information at various layers of a protocal stack

PACKET SNIFFER:-

*) Capture Packets sent and recieves by the computer
*) Displays protocal information of capture Pockets.
*) Works possessively
*) Recieves copies of networks packet for analysis

PACKET SNIFFER TOOLS:-

*) TCP dump - Command line packet Capture tool
*) Wiresshark - Graphical packet analyzer

Wireshark · Display Filters

Filter Name	Filter Expression
Ethernet address 00:00:5e:00:53:00	eth.addr == 00:00:5e:00:53:00
Ethernet type 0x0806 (ARP)	eth.type == 0x0806
Ethernet broadcast	eth.addr == ff:ff:ff:ff:ff:ff
No ARP	not arp
IPv4 only	ip
IPv4 address 192.0.2.1	ip.addr == 192.0.2.1
IPv4 address isn't 192.0.2.1	ip.addr != 192.0.2.1
IPv6 only	ipv6
IPv6 address 2001:db8::1	ipv6.addr == 2001:db8::1
TCP only	tcp
UDP only	udp
Non-DNS port	!(udp.port == 53 || tcp.port == 53)
TCP or UDP port is 80 (HTTP)	tcp.port == 80 || udp.port == 80
HTTP	http
No ARP and no DNS	not arp and not dns
Non-HTTP and non-SMTP to/from 192.0.2.1	ip.addr == 192.0.2.1 and tcp.port not in {80, 25}

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OK Cancel Help

DESCRIPTION:-

Wireshark is a networks protocals analyzer that capture and displays network packets is real time It helps monitor, analyze, troubleshoot and inspect network traffic. Using filters and packet details.

USES OF NETWORKS:-

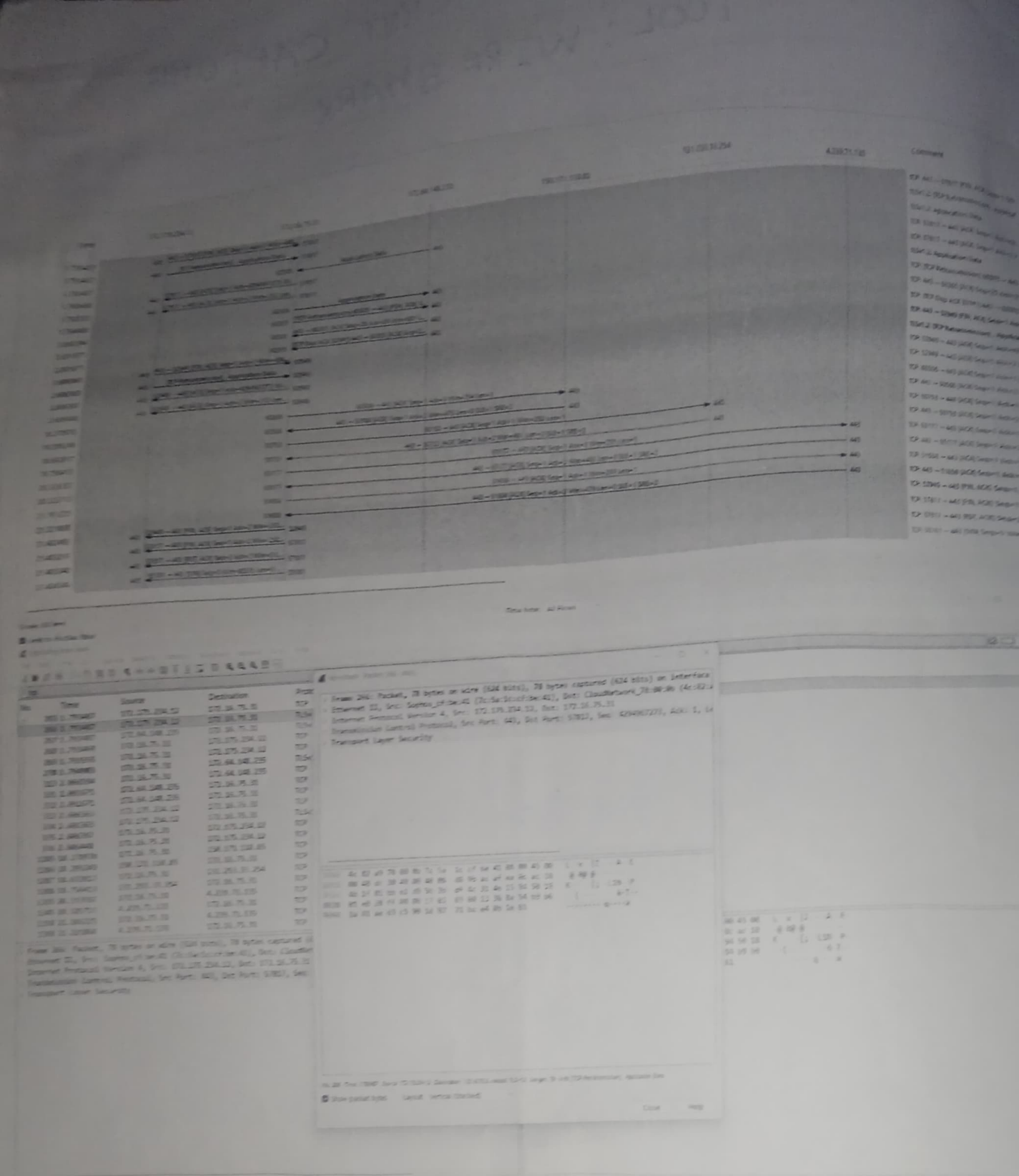
*) Capture networks traffics
*) Analyzes Protocal
*) Applies displays filter
*) Monitors Networks issues

APPLICATIONS:-

*) Network troubleshooting
*) Security analysis
*) Protocal debugging
*) Learning network protocals

CAPTURING PACKET:-

1. Open wireshark
2. Select the network interface
3. Click start captures
4. Observe Packet in real time
5. Click Stop and Save the capture



4. Create a filter to display only IP/ICMP Packets

1. Open the capture file
2. Apply the IP or ICMP filter
3. Inspect and view the flow graph and save capture.

5. Create a filter to display only DHCP Packet

1. Open the capture
2. Apply the ~~dhcp~~ dhcp file
3. Inspect and view the flow graph and save the capture.

RESULT:-

Thus the features of wireshark and the implement of different filter was Successfully Complied.